Assignment7 – Learning

Given: June 3 Due: June 8

Problem 7.1 (Decision List)

We want to construct a decision list to classify the data below where result values V depend on 4 attributes A, B, C, D. The tests should be conjunctions of literals.

1. Assume your literals must be of the form $attribute = number$. Which values of k allow for giving the shortest possible decision list in k-DL (i.e., using at most k literals per test)? Give one such list.

Solution: For $k = 3$ (or higher), we can build a list of length 2:

if $A = 0 \wedge C = 1 \wedge D = 0$ then 0 else
if $A = 0 \wedge C = 0 \wedge D = 1$ then 0 else
1.

Technically not correct but also interesting is the following solution for $k = 2$ with a list of length 3:

if $A = 1$ then 1 else
if $C = 0 \wedge D = 0$ then 1 else
if $C = 1 \wedge D = 1$ then 1 else
0

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2. Now assume your literals may also be of the form $attribute = attribute$. Answer the same question as above.

Solution: For $k = 1$, we can build a list of length 2:

if $A = 1$ then 1 else
if $C = D$ then 1 else
0

Example	A	B	C	D	V
#1	1	0	0	0	1
#2	1	0	1	1	1
#3	0	1	0	0	1
#4	1	1	0	1	1
#5	0	0	1	1	1
#6	0	1	1	0	0
#7	0	1	0	1	0
#8	0	0	1	0	0

else

else a=0, c=1, d=1, b=0

a=0, c= 0 ,d=1
a=0, c=1, d=0

Problem 7.2 (General Properties of Linear Regression)

Consider a list of examples $(\vec{x}_i, y_i) \in \mathbb{R}^n \times \mathbb{R}$ for $i = 1, \ldots, m$. We want to apply linear regression.

m: number of training examples.
n: number of input features.
xi : input vector of example i.
yi : correct numerical output for example i.

w1,…,w n are weights,

w0 is the constant term or bias,

h(x) is the predicted output.

1. What is the hypothesis space?

Solution: The set of linear functions $\vec{x} \mapsto \vec{w} \cdot \vec{x} + w_0$.

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2. What is the point of the trick to set $(x_i)_0 = 1$ for all examples?

Solution: If we assume all x_i are of the form $\langle 1, (\vec{x}_i)_1, \ldots, (\vec{x}_i)_m \rangle$ we can write the hypotheses as functions $\vec{x} \mapsto \vec{w} \cdot \vec{x}$ for vectors $\vec{w} \in \mathbb{R}^{1+n}$. That simplifies the calculations. tldr: 1 is better for calculation than starting with initial 0

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3. What is the maximum number of examples for which a consistent hypothesis can still exist?

Solution: Trick question: There can be infinitely many $\vec{x}$. So if we only pick examples of the form $(\vec{x}, h(\vec{x}))$ for some fixed hypothesis h, we can find infinitely many examples and still have h as a consistent hypothesis.
But we do have the following: If the examples $(\vec{x}, y)$ are chosen randomly and IID, then requiring consistency, i.e., $\vec{w} \cdot \vec{x}_i = y_i$ for $i = 1, \ldots, m$, yields a system of m linear equations in $1 + n$ unknowns (the components of $\vec{w}$). So we can realistically expect being able to find a consistent hypothesis only for up to $1 + n$ examples. ✱ n normal weights plus the bias w0

independent and identically distributed

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4. How important is it to find a consistent hypothesis here?

Solution: With continuous data (real numbers instead of, e.g., Boolean classifications), a very limited hypothesis space (only linear functions instead of all functions), and many examples (typically much more than n), it is unlikely that any hypothesis is consistent. Therefore, practical applications aim for error minimization instead of consistency. A consistent hypothesis would have error 0. Error minimization can be seen as finding the least inconsistent hypothesis.

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5. What kind of loss function should we use here?

Solution: $L_{0/1}$ loss (error rate) is not a good choice. It just counts how often $h(\vec{x}_i) \neq y_i$, but equality comparisons on real number data are typically not reliable.
Instead, we should a loss function that measures how far off a hypothesis is. Typical choices are $|h(\vec{x}_i) - y_i|$ and $(h(\vec{x}_i) - y_i)^2$.

Problem 7.3 (Support Vectors)

Consider the following 2-dimensional dataset

support vector	classification
$\mathbf{x}_1 = \langle 0, 0 \rangle$	$\mathbf{y}_1 = -1$
$\mathbf{x}_2 = \langle 0, 0.5 \rangle$	$\mathbf{y}_2 = -1$
$\mathbf{x}_3 = \langle 0.5, 0 \rangle$	$\mathbf{y}_3 = -1$
$\mathbf{x}_4 = \langle 1, 1 \rangle$	$\mathbf{y}_4 = 1$
$\mathbf{x}_5 = \langle 2, 2 \rangle$	$\mathbf{y}_5 = -1$

1. Give a linear separator in the form $h(\mathbf{x}) = \mathbf{w} \cdot \mathbf{x} + b$ for the dataset containing only the examples for $\mathbf{x}_1$ to $\mathbf{x}_4$.

   Solution: Many solutions, e.g., $\mathbf{w} = \langle 1, 1 \rangle$ and $b = -1$.

2. Explain informally why no linear separator exists for the full dataset of all 5 vectors.

   Solution: The points $\mathbf{x}_1, \mathbf{x}_4, \mathbf{x}_5$ lie on a line and the middle one has a different classification than the others. No line can have $\mathbf{x}_1$ and $\mathbf{x}_5$ on one side and $\mathbf{x}_4$ on the other.

3. Transform the dataset into a 3-dimensional dataset by applying the function $F(\langle u, v \rangle) = \langle u^2, v^2, u + v \rangle$.

   Solution:

support vector $\mathbf{x}$	$F(\mathbf{x})$	classification
$\mathbf{x}_1$	$\langle 0, 0, 0 \rangle$	$\mathbf{y}_1 = -1$
$\mathbf{x}_2$	$\langle 0, 0.25, 0.5 \rangle$	$\mathbf{y}_2 = -1$
$\mathbf{x}_3$	$\langle 0.25, 0, 0.5 \rangle$	$\mathbf{y}_3 = -1$
$\mathbf{x}_4$	$\langle 1, 1, 2 \rangle$	$\mathbf{y}_4 = 1$
$\mathbf{x}_5$	$\langle 4, 4, 4 \rangle$	$\mathbf{y}_5 = -1$

4. Give a linear separator for the transformed full dataset in the form $h(\mathbf{x}) = \mathbf{w} \cdot \mathbf{x} + b$.

   Solution: Many solutions, e.g., $\mathbf{w} = \langle -1, -1, 2 \rangle$ and $b = -1$.

Problem 7.4 (Weight Updates)

We're trying to find a linear separator. Our examples are the set

Example number	$\mathbf{x}_1$	$\mathbf{x}_2$	y
1	2	0	2
2	3	1	2

Our hypothesis space contains the functions $h_{\mathbf{w}}(\mathbf{x}) = A(\mathbf{w}\cdot\mathbf{x})$ for 2+1-dimensional vectors $\mathbf{w}, \mathbf{x}$ (using the trick $\mathbf{x}_0 = 1$ to allow for the constant term $\mathbf{w}_0$) and some fixed function A.

As the initial weights, we use $\mathbf{w}_0 = \mathbf{w}_1 = \mathbf{w}_2 = 0$.

For each of the following cases, iterate the respective weight update rule once for each example (using the examples in the order listed). Use learning rate $\alpha = 1$.

1. Using the threshold function $A(z) = \mathcal{T}(z)$, i.e., $A(z) = 1$ if $z > 0$ and $A(z) = 0$ otherwise. Here we cannot do gradient descent, so we have to use the perceptron learning rule.

Solution: The update rule is $\mathbf{w}_i \leftarrow \mathbf{w}_i + \alpha(y - h_{\mathbf{w}}(\mathbf{x}))\mathbf{x}_i$. Using the examples, we obtain:

- Example 1: $y - h_{\mathbf{w}}(\mathbf{x}) = 2 - \mathcal{T}((0,0,0)\cdot(1,2,0)) = 2$, i.e., $\mathbf{w}_i \leftarrow \mathbf{w}_i + 1\mathbf{x}_i$. Thus, $\mathbf{w} \leftarrow (2,4,0)$.
- Example 2: $y - h_{\mathbf{w}}(\mathbf{x}) = 2 - \mathcal{T}((2,4,0)\cdot(1,3,1)) = 1$, i.e., $\mathbf{w}_i \leftarrow \mathbf{w}_i + 1\mathbf{x}_i$. Thus, $\mathbf{w} \leftarrow (3,7,1)$.

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2. Using the logistic function $A(z) = 1/(1+e^{-x})$. Here we use gradient descent.

Solution: The update rule is $\mathbf{w}_i \leftarrow \mathbf{w}_i + \alpha(y - h_{\mathbf{w}}(\mathbf{x}))h_{\mathbf{w}}(\mathbf{x})(1 - h_{\mathbf{w}}(\mathbf{x}))\mathbf{x}_i$. Using the examples, we obtain:

- Example 1: $h_{\mathbf{w}}(\mathbf{x}) = 1/(1 + e^0) = 1/2$, i.e., $\mathbf{w}_i \leftarrow \mathbf{w}_i + 1/4(2 - 1/2)\mathbf{x}_i$. Thus, $\mathbf{w} \leftarrow (3/8, 3/4, 0)$.
- Example 2: $h_{\mathbf{w}}(\mathbf{x}) = 1/(1 + e^{-((3/8,3/4,0)\cdot(1,3,1))}) = 1/(1 + e^{-21/8})$, i.e., (after rounding) $\mathbf{w}_i \leftarrow \mathbf{w}_i + 0.07\mathbf{x}_i$. Thus, $\mathbf{w} \leftarrow (0.44, 0.95, 0.07)$.

Problem 7.5 (XOR Neural Network)

Consider the following neural network with

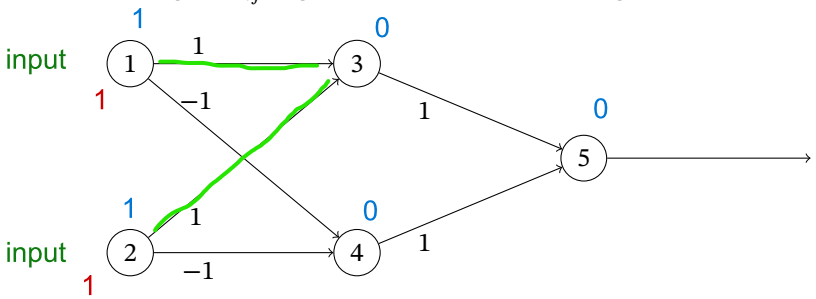
- inputs a_1 and a_2
- units 3, 4, 5 with activation functions such that $a_i \leftarrow \begin{cases} 1 & \text{if } \Sigma_j w_{ji} a_j > b_i \\ 0 & \text{otherwise} \end{cases}$

a3= (1*1)+(1*1) = 2 > (b = 0) => 1

a4 = (1*-1) + (1*-1) = -2 < (b=0) => 0 ie. otherwise

a5 = (1*1) + (1 * 0) = 1 > (b=0) => 1

- weights w_{ij} as given by the labels on the edges



bias in blue

1. Assume $b_3 = b_4 = b_5 = 0$ and inputs $a_1 = a_2 = 1$. What are the resulting activations a_3, a_4, and a_5?

Solution: $a_3 = 1, a_4 = 0, a_5 = 1$

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2. Choose appropriate values for b_3, b_4, and b_5 such that the network implements the XOR function.

Solution: E.g., $b_3 = 0.5$, $b_4 = -1.5$, $b_5 = 1.5$. More generally, any values work that satisfy $0 \leq b3 < 1$, $-2 \leq b4 < -1$, and $1 \leq b5 < 2$.

A	B	X (A ⊕ B)
0	0	0
0	1	1
1	0	1
1	1	0